# Machine Learning Prediction of Bulk Modulus from Crystal-Structure-Derived Features

## A preliminary computational materials-ML study


## Abstract

This project investigates whether machine-learning models can predict the bulk modulus of crystalline materials from structure-derived and composition-based descriptors. The Matbench matbench_log_kvrh dataset contains 10,987 materials and uses log10(K_VRH), the base-10 logarithm of the Voigt–Reuss–Hill bulk modulus in GPa, as the target. Density-related features were generated using Matminer's DensityFeatures, and elemental descriptors were generated using the Magpie preset. Linear Regression and a Random Forest regressor were trained using an 80/20 random train/test split. Random Forest achieved MAE = 0.0708, RMSE = 0.1239, and $R^2$ = 0.8930 on the held-out test set, outperforming Linear Regression. The most important feature was volume per atom (vpa), followed by mean melting temperature and mean Np valence. These results are preliminary and should not be treated as validation for industrial material selection.


## Introduction and motivation

Bulk modulus quantifies a material's resistance to uniform compression. It is one of several properties that engineers may consider when screening materials for pressure-bearing components and industrial equipment. A data-driven model that estimates bulk modulus from material descriptors may help explore relationships between composition, structure, and mechanical response. However, bulk modulus alone does not establish suitability for a specific application. Yield strength, fracture toughness, fatigue, creep, corrosion or oxidation resistance, thermal stability, and operating conditions may also be critical.

## Research question

How accurately can machine-learning models predict the bulk modulus of crystalline materials using crystal-structure-derived features, and which features contribute most to their predictions?

## Dataset and target

The notebook loads the Matbench matbench_log_kvrh dataset through Matminer's load_dataset function. The dataset contains 10,987 records and three original columns, including the crystal structure and target. The target, log10(K_VRH), represents the base-10 logarithm of the Voigt–Reuss–Hill bulk modulus expressed in GPa. The notebook converts this target back to GPa for exploratory descriptive statistics, but uses the logarithmic target for model training and evaluation.

| Statistic (GPa) | Value |
|---|---|
| Count | 10,987 |
| Mean | 100.85 |
| Standard deviation | 68.14 |
| Minimum | 1 |

| | |
|---|---|
| 25th percentile | 47 |
| Median | 86 |
| 75th percentile | 144 |
| Maximum | 575 |

## Feature engineering and data preparation

The notebook generated three structure-related descriptors using Matminer's DensityFeatures: density, volume per atom (vpa), and packing fraction. It then derived each material's composition from its crystal structure and used ElementProperty.from_preset('magpie') to generate composition-based elemental descriptors. The resulting modeling table contained 139 columns, of which 135 numeric columns were used as input features. The structure, composition, target, and derived GPa-scale target were excluded from the feature matrix.

The target had no missing values, and the notebook reported no missing values in the training or test feature matrices. The data were randomly divided into 8,789 training records (80%) and 2,198 test records (20%), using random_state=42. Median imputation was included in both modeling pipelines. Linear Regression also used StandardScaler; the Random Forest model used 100 trees and random_state=42.

## Models and evaluation

Linear Regression was used as a baseline and compared with a Random Forest regressor. Performance was measured using mean absolute error (MAE), root mean squared error (RMSE), and the coefficient of determination ($R^2$). Because the models predict log10(K_VRH), MAE and RMSE are reported on the logarithmic target scale and should not be interpreted as errors measured directly in GPa.

| Metric | Linear Regression | Random Forest | Better result |
|---|---|---|---|
| MAE | 0.0985 | 0.0708 | Random Forest (lower) |
| RMSE | 0.1542 | 0.1239 | Random Forest (lower) |
| $R^2$ | 0.8343 | 0.8930 | Random Forest (higher) |

On this test split, Random Forest reduced MAE by approximately 28.1% and RMSE by approximately 19.7% relative to Linear Regression. $R^2$ increased by about 0.0588. This indicates that Random Forest performed better on the selected held-out sample; it does not establish that the same performance will generalize to all material families or experimental measurements.

## Feature-importance analysis

The notebook's Random Forest feature-importance output ranked the following features highest. The table below reproduces the values from the notebook rather than the earlier, incorrect list.

| Rank | Feature | Importance |
|---|---|---|
| 1 | vpa | 0.4178 |
| 2 | MagpieData mean MeltingT | 0.1468 |
| 3 | MagpieData mean NpValence | 0.0604 |
| 4 | density | 0.0544 |
| 5 | packing fraction | 0.0380 |
| 6 | MagpieData mean NUnfilled | 0.0301 |
| 7 | MagpieData maximum MendeleevNumber | 0.0218 |
| 8 | MagpieData mean Row | 0.0167 |
| 9 | MagpieData mean Electronegativity | 0.0108 |
| 10 | MagpieData maximum GSvolume_pa | 0.0101 |

Volume per atom (vpa) had the highest importance score (0.4178). Mean melting temperature and mean Np valence ranked second and third. The ranking suggests that volume-related structural information and selected elemental descriptors were useful to this model. These importance scores are impurity-based and indicate how the fitted Random Forest used the features; they do not demonstrate physical causation. Correlated descriptors can also influence rankings.

## Discussion and limitations

- The Random Forest model outperformed Linear Regression across all three reported test metrics, suggesting that it captured patterns in this dataset that the linear baseline did not capture as effectively.
- The predicted-versus-actual plot should be interpreted as a visual check of agreement on the test split. Points closer to the diagonal represent closer predictions; deviations indicate prediction error.
- The target data are computationally derived in Matbench, rather than direct experimental measurements.
- A random split can place similar materials in both training and test sets. Strong performance on this split does not guarantee performance on entirely new chemical or structural families.
- The study compares only two models and does not include extensive hyperparameter tuning, cross-validation, uncertainty quantification, or external validation.
- Bulk modulus is only one aspect of mechanical performance and cannot independently determine suitability for harsh environments or industrial pressure service.

## Conclusion and next steps

This preliminary study demonstrates a reproducible workflow for generating structure- and composition-based descriptors and comparing machine-learning regressors for bulk-modulus prediction. Random Forest achieved MAE = 0.0708, RMSE = 0.1239, and $R^2 = 0.8930$ on the notebook's 20% test split, outperforming Linear Regression. The feature-importance analysis identified volume per atom as the leading predictor, followed by mean melting temperature and mean Np valence. Future work should evaluate cross-validation and material-family-based splits, tune hyperparameters, compare additional models, examine prediction uncertainty, and validate predictions against independent or experimental data before considering engineering use.

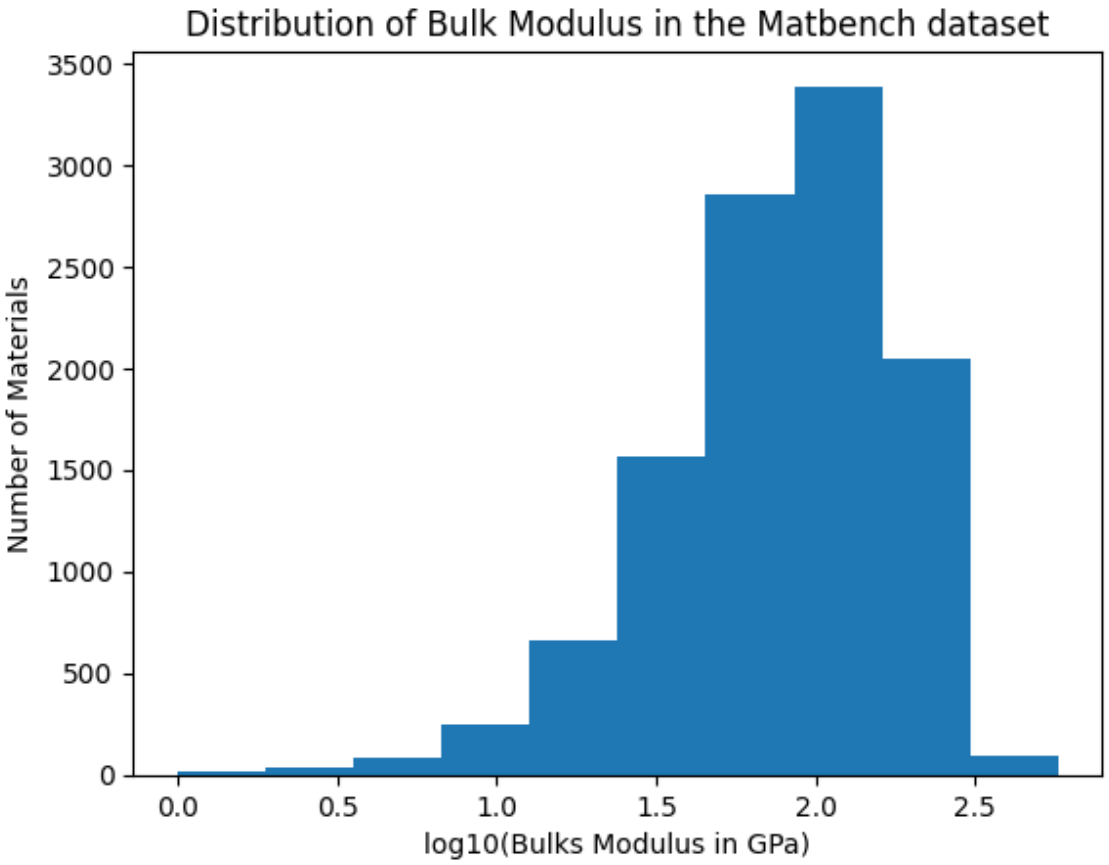
## References and resources

- Matbench repository and benchmark information: https://github.com/materialsproject/matbench
- Matminer dataset summary: https://hackingmaterials.lbl.gov/matminer/dataset_summary.html
- Matbench log_kvrh dataset record: https://figshare.com/articles/dataset/log_kvrh/9755477
- Matminer bulk-modulus example: https://hackingmaterials.lbl.gov/matminer/example_bulkmod.html
- scikit-learn documentation: https://scikit-learn.org/stable/

## Figures generated by the notebook

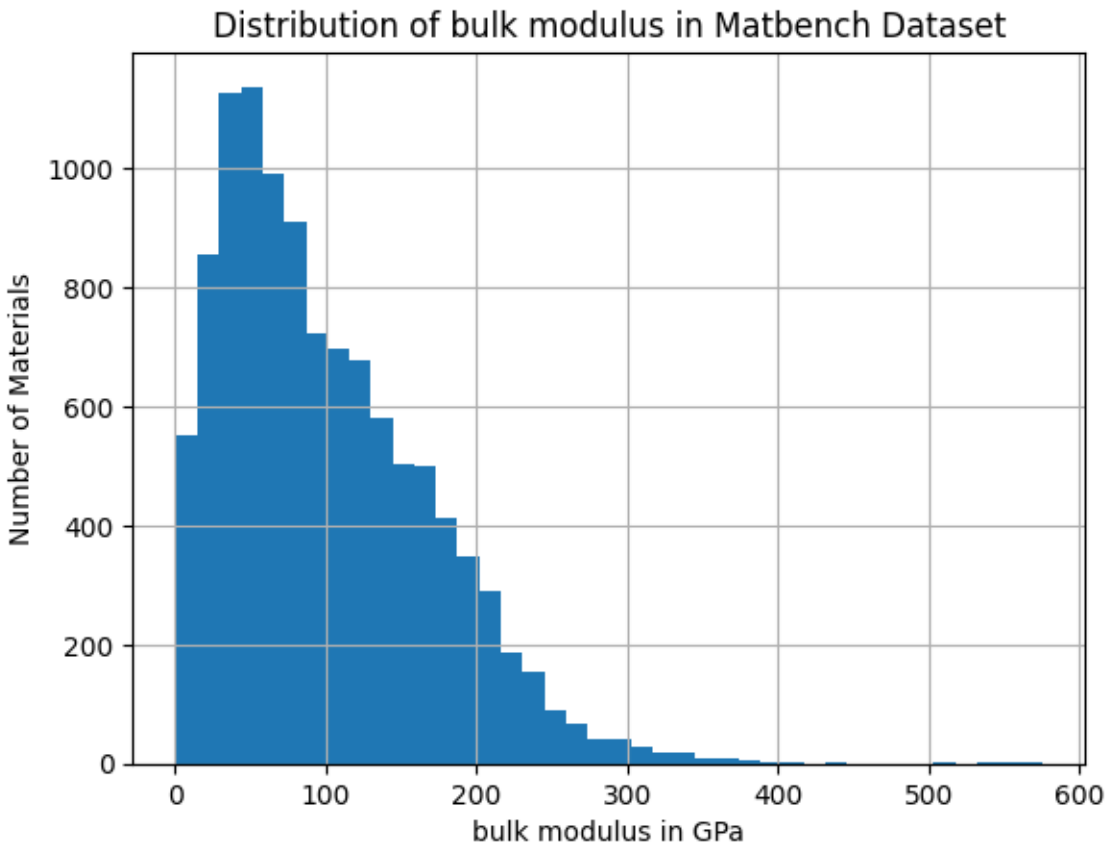
The exploratory histograms, Random Forest feature-importance plot, and predicted-versus-actual scatter plot below are embedded from the saved outputs in the original notebook. The two model-comparison plots are additional figures; matching plotting code has been added to the polished notebook so they can be regenerated from its calculated evaluation results after a clean run in Google Colab.

**Figure 1. Distribution of log10(K_VRH)**



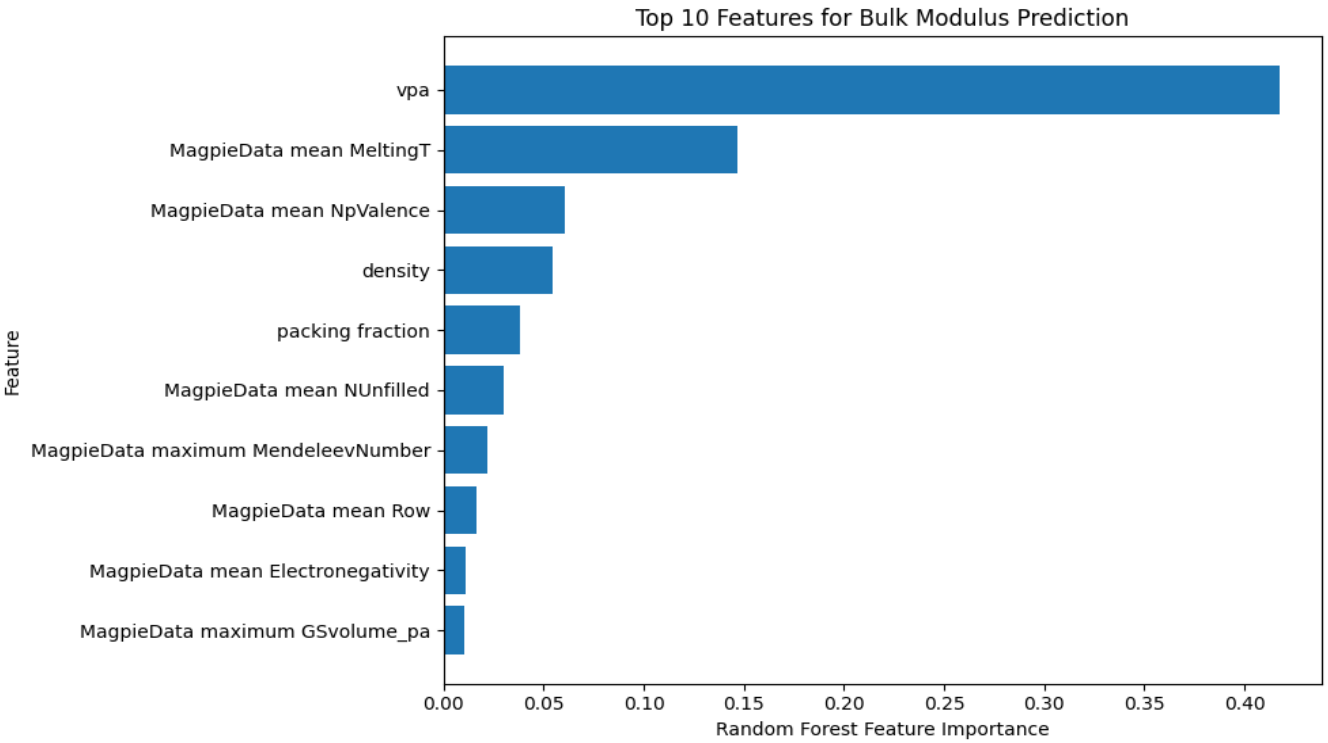

The target distribution on the logarithmic scale used for model training.

**Figure 2. Distribution of bulk modulus in GPa**



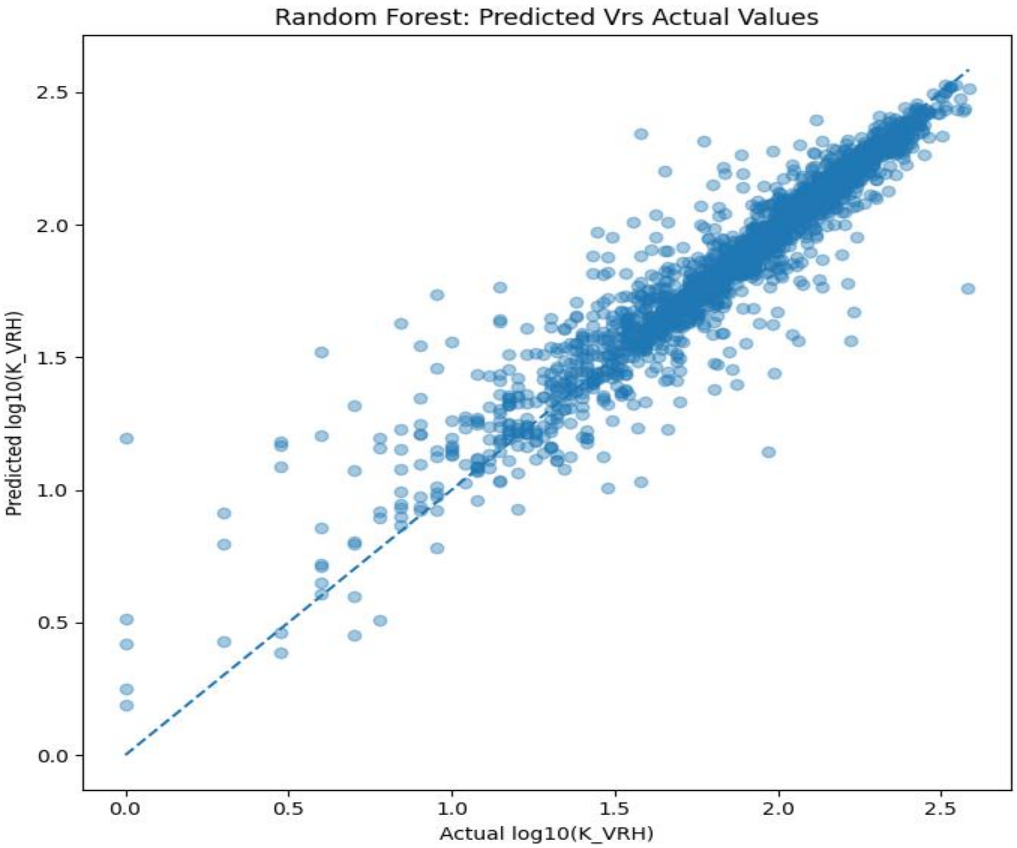

The target transformed back to GPa for descriptive exploration; the distribution is right-skewed.

**Figure 3. Top 10 Random Forest feature importances**



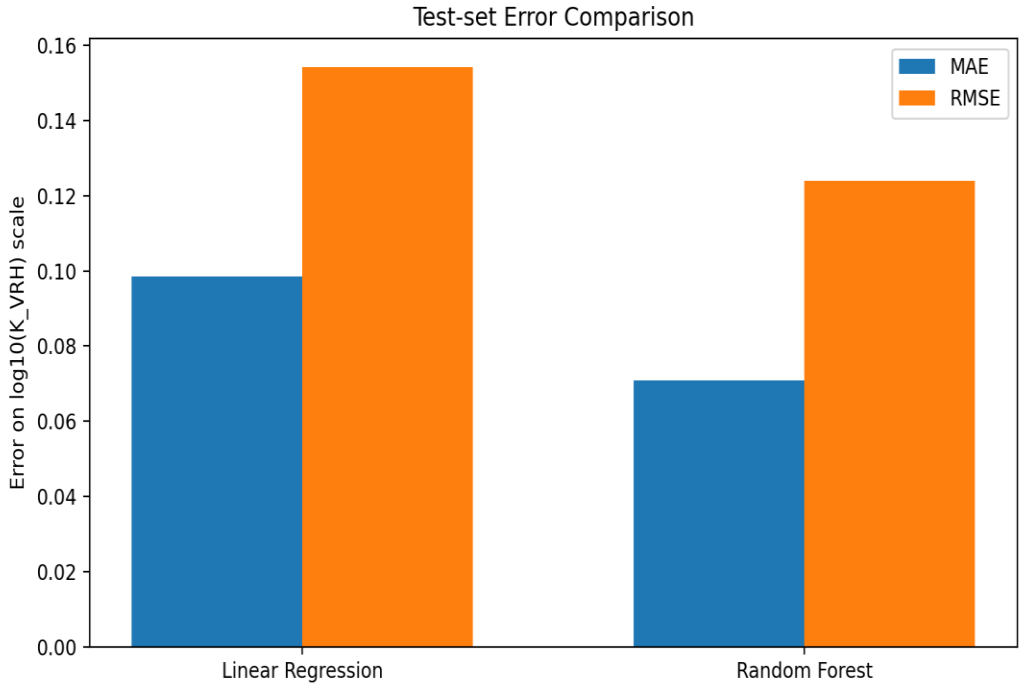

The model's impurity-based feature importance scores; these are predictive associations, not causal effects.

**Figure 4. Predicted versus actual log10(K_VRH)**



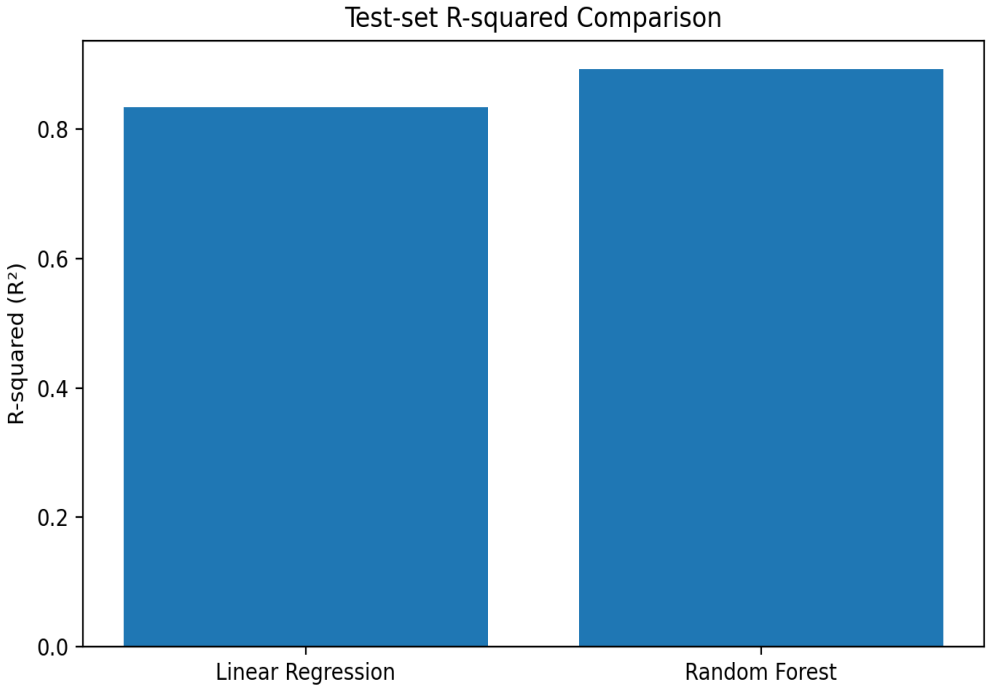

Each point compares a Random Forest prediction with the corresponding held-out test target; the diagonal indicates perfect agreement.

**Figure 5. Test-set error comparison**




MAE and RMSE for both models on the log10(K_VRH) scale. Lower values indicate smaller prediction errors.

**Figure 6. Test-set R-squared comparison**




$R^2$ for the two regression models on the held-out test set. Random Forest has the higher $R^2$ in this split.

Consistency notes: Figures 1–4 are original saved plot outputs from the notebook. Figures 5–6 are newly added to both the report and notebook; the notebook code regenerates them from the evaluation results. Re-run the polished notebook from top to bottom before final submission and verify the metric values remain the same.